Lab no. 3

Working with Git History, Stashing, and Reverting Commits

NAME: **LAIBA RANI**

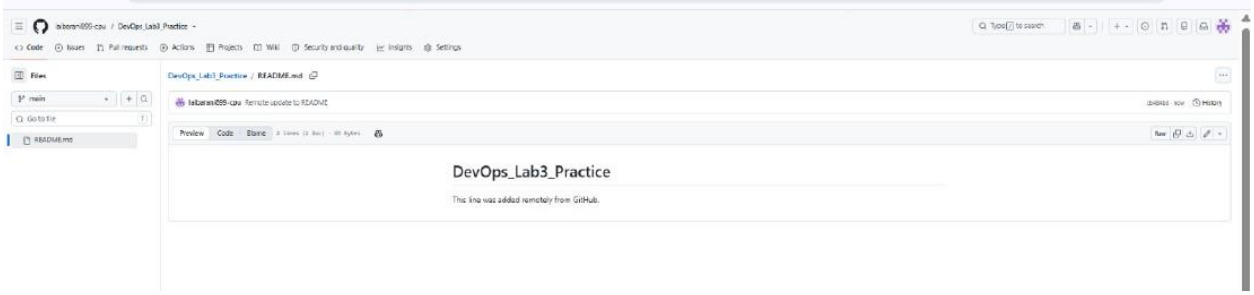
REGISTRATION NO.: **2023-BCS-053**

COURSE CODE: **CS-649**

SECTION: **A**

GITHUB USERNAME: **laibarani899-cpu**

Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull—rebase)



README.md edited directly on GitHub (commit “Remote update to README”).

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local update to README"
[main 88ff31e] Local update to README
 1 file changed, 1 insertion(+)
```

Local change to README.md staged and committed (“Local update to README”).

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'github.com:laibarani899-cpu/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

git push rejected (non-fast-forward) because the remote has commits that are not in the local branch.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git pull --no-rebase origin main
From github.com:laibarani899-cpu/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
fatal: refusing to merge unrelated histories
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git pull --no-rebase origin main --allow-unrelated-histories
From github.com:laibarani899-cpu/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
Auto-merging README.md
CONFLICT (add/add): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch and 'origin/main' have diverged,
and have 7 and 2 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)

All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

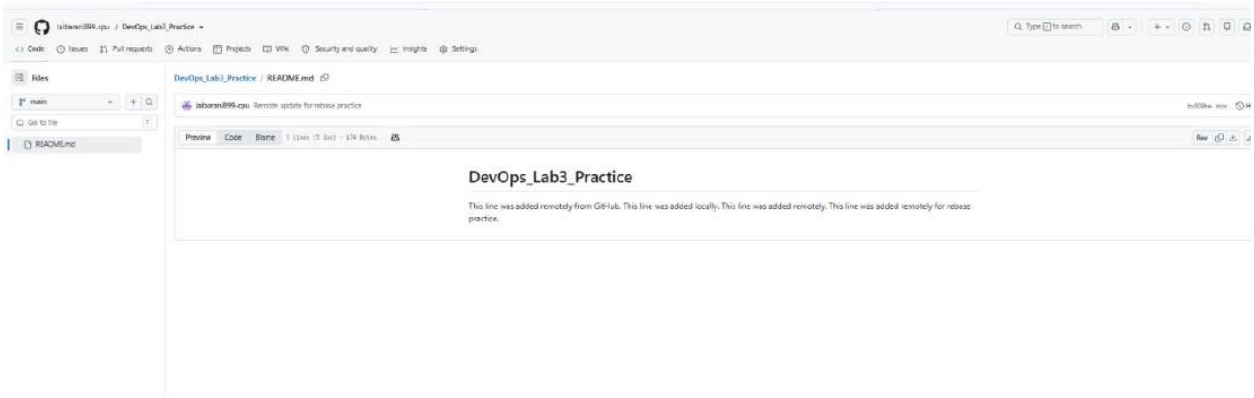
Changes to be committed:
        modified:   README.md

laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Merge remote changes into README"
[main 077166b] Merge remote changes into README
```

git pull—no-rebase (retried with—allow-unrelated-histories); conflict in README.md fixed manually and merge commit created.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 2 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (29/29), 3.71 KiB | 253.00 KiB/s, done.
Total 29 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   6c1fe05..077166b  main -> main
```

Successful git push origin main after the merge.



Second remote edit made on GitHub for the rebase exercise ("Remote update for rebase practice").

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 978 bytes | 108.00 KiB/s, done.
From github.com:laibarani899-cpu/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   077166b..bc809be  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 375c158... Local update for rebase practice
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"

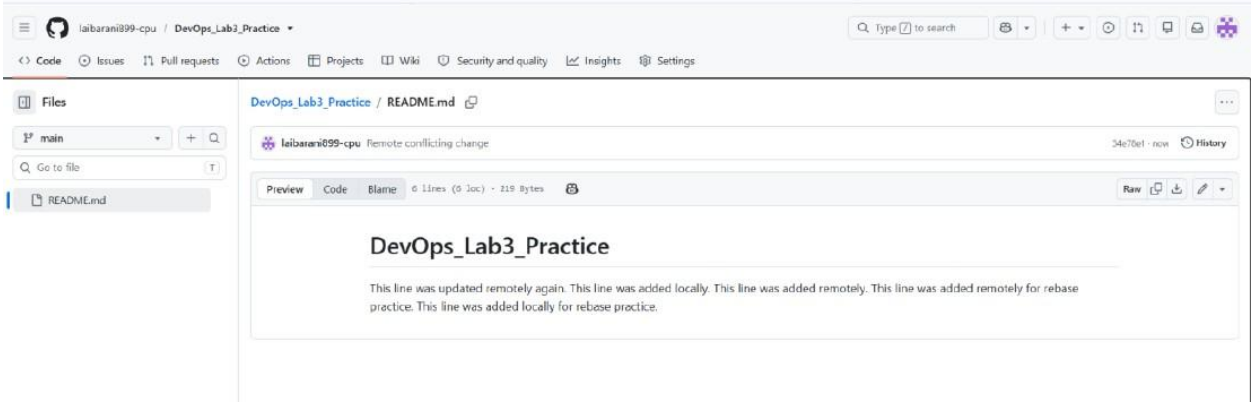

laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD ce9c922] Local update for rebase practice
 1 file changed, 1 insertion(+)
Successfully rebased and updated refs/heads/main.
```

git pull—rebase origin main replays the local commit on top of the remote commit; conflict resolved, then git add and git rebase—continue.

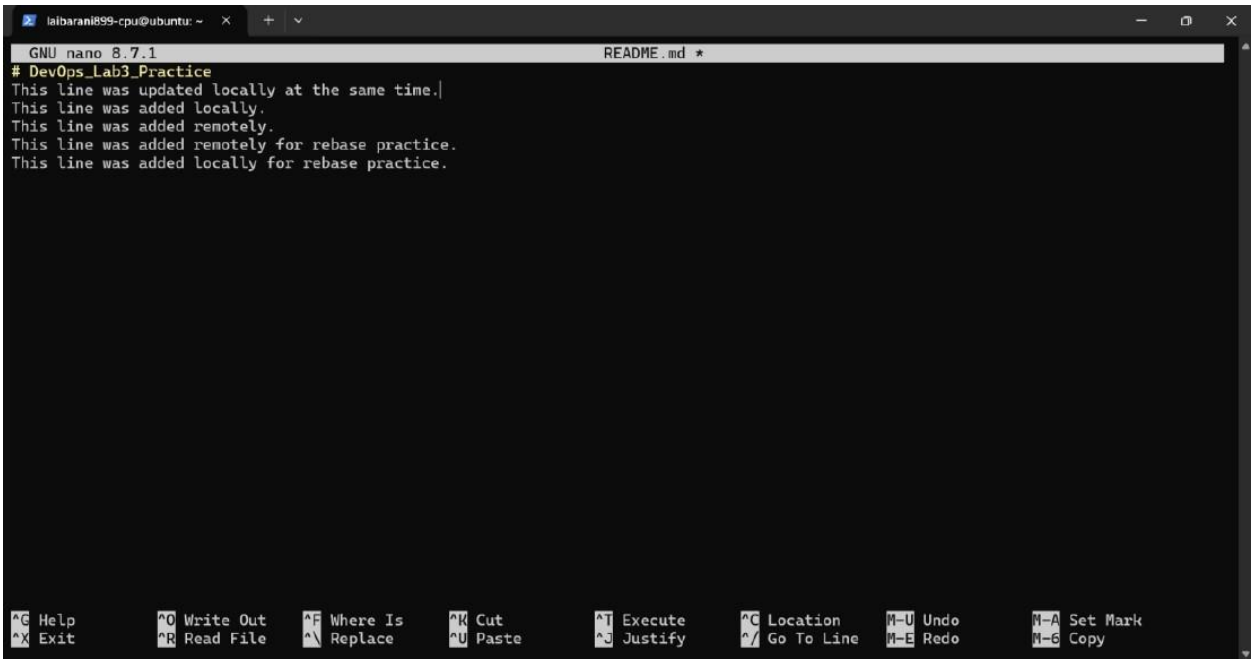
```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 344 bytes | 68.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   bc809be..ce9c922  main -> main
```

Successful push after the rebase, keeping the history linear.

Task 2: Creating and Resolving Merge Conflicts Manually



Conflicting change committed on GitHub ("Remote conflicting change").



Same line of README.md edited locally in nano.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local conflicting change"
[main d4b7aca] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Local change staged and committed (“Local conflicting change”).

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

git push rejected (fetch first) because of the conflicting remote commit.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
Unpacking objects: 100% (3/3), 975 bytes | 487.00 KiB/s, done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
From github.com:laibarani899-cpu/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   ce9c922..34e78e1  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply d4b7aca... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply d4b7aca... # Local conflicting change
```

git pull—rebase pauses with a merge conflict in README.md.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice
This line was updated remotely again.
This line was updated locally at the same time.
This line was added locally.
This line was added remotely.
This line was added remotely for rebase practice.
This line was added locally for rebase practice.
```

README.md after manually resolving the conflict (both changes kept).

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD 6c2a840] Local conflicting change
 1 file changed, 1 insertion(+)
Successfully rebased and updated refs/heads/main.
```

Conflict marked resolved with git add, then git rebase—continue completes the rebase.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 349 bytes | 116.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   34e78e1..6c2a840  main -> main
branch 'main' set up to track 'origin/main'.
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$
```

Resolved changes pushed to the remote repository.

Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ mkdir textfiles
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ echo "File A content" > textfiles/a.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ echo "File B content" > textfiles/b.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ echo "File C content" > textfiles/c.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ ls textfiles
a.txt  b.txt  c.txt
```

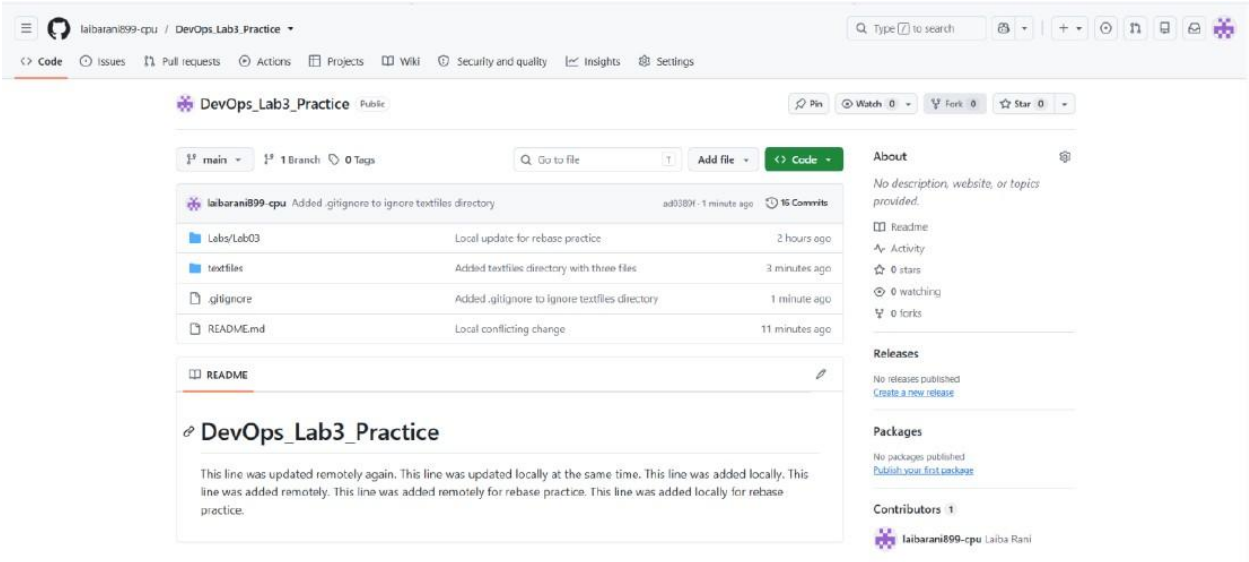
textfiles directory created with a.txt, b.txt and c.txt.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add .
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added textfiles directory with three files"
[main d182ea9] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 491 bytes | 40.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   6c2a840..d182ea9  main -> main
```

textfiles directory committed and pushed to GitHub.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ echo "textfiles/*" > .gitignore
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ cat .gitignore
textfiles/*
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add .gitignore
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added .gitignore to ignore textfiles directory"
[main ad0389f] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 379 bytes | 189.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   d182ea9..ad0389f  main -> main
```

.gitignore created with textfiles/*, then committed and pushed.



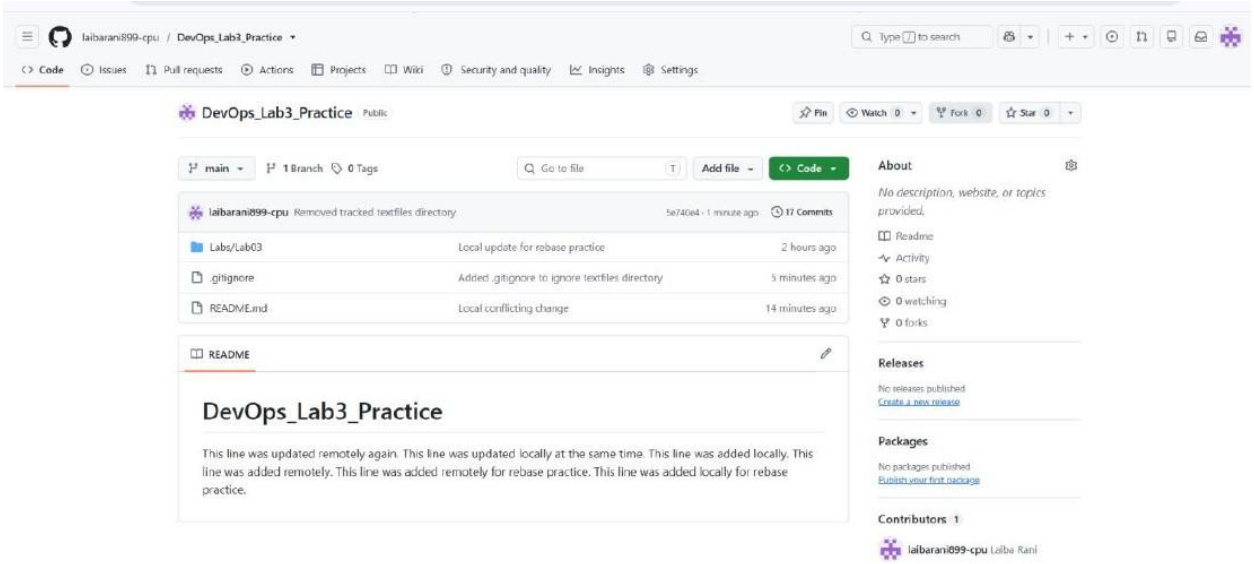
GitHub repository still shows the textfiles directory after adding .gitignore.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        deleted:    textfiles/a.txt
        deleted:    textfiles/b.txt
        deleted:    textfiles/c.txt

laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add .
git commit -m "Removed tracked textfiles directory"
[main 5e740e4] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 237 bytes | 26.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   ad0389f..5e740e4  main -> main
```

git rm -r—cached textfiles, then commit and push (files stay on the local disk).



GitHub repository after the push; the textfiles directory is removed from the remote.

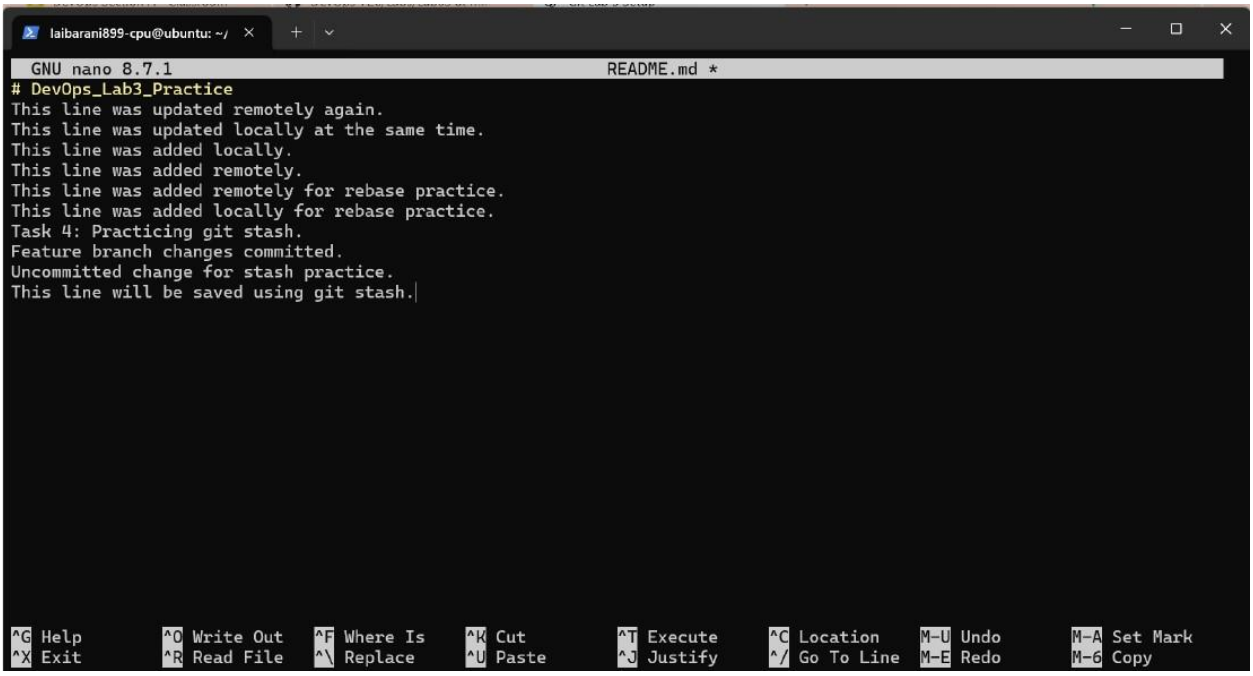
Task 4: Create Temporary Changes and Use git stash



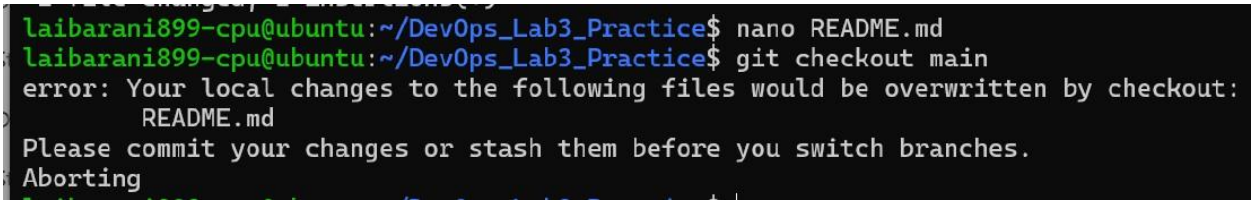
README.md edited on feature-branch.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b feature-bran
ch
Switched to a new branch 'feature-branch'
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
git commit -m "Changes the README.md file"
[feature-branch 5fc0eb2] Changes the README.md file
 1 file changed, 2 insertions(+)
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$
```

feature-branch created and the README.md change committed (“Changes the README.md file”).



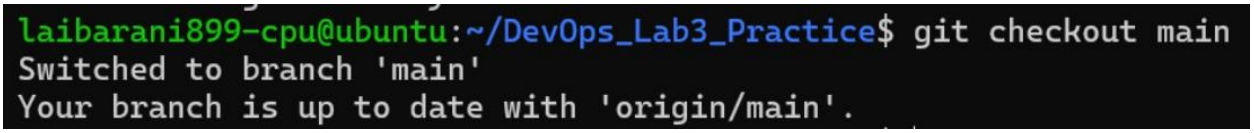
New uncommitted lines added to README.md.



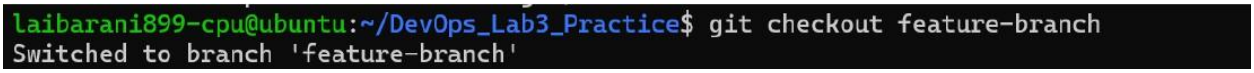
git checkout main fails because the local changes would be overwritten.



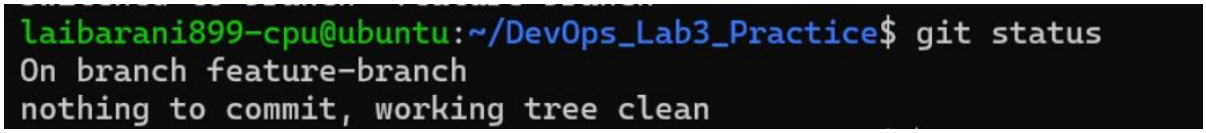
Uncommitted changes saved with git stash.



Successfully switched to the main branch after stashing.



Switched back to feature-branch.



git status shows nothing to commit, working tree clean.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (8cdde94aca5063d7666e6f2a7fb52d306e1fa345)
```

git stash pop restores the stashed changes in README.md.

Task 5: Checkout a Specific Commit Using git log

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
5fc0eb2 (HEAD -> feature-branch) Changes the README.md file
5e740e4 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
ad0389f Added .gitignore to ignore textfiles directory
d182ea9 Added textfiles directory with three files
6c2a840 Local conflicting change
34e78e1 Remote conflicting change
ce9c922 Local update for rebase practice
bc809be Remote update for rebase practice
077166b Merge remote changes into README
88ff31e Local update to README
6c1fe05 Remote update to README
eff6819 Initial commit
7374766 Local update for rebase practice
35ca750 Merge branch 'main' of github.com:laibarani899-cpu/DevOps_Lab3_Practice
eaa7517 Local update to README
cb484b3 Remote update to README
71e95a3 Initial commit
06cdfdd Set up Lab 3 practice structure
```

Commit history shown with git log—oneline.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git checkout ad0389f
Note: switching to 'ad0389f'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at ad0389f Added .gitignore to ignore textfiles directory
```

git checkout ad0389f puts the repository in detached HEAD state.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Previous HEAD position was ad0389f Added .gitignore to ignore textfiles directory
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

Returned to the main branch with git checkout main.

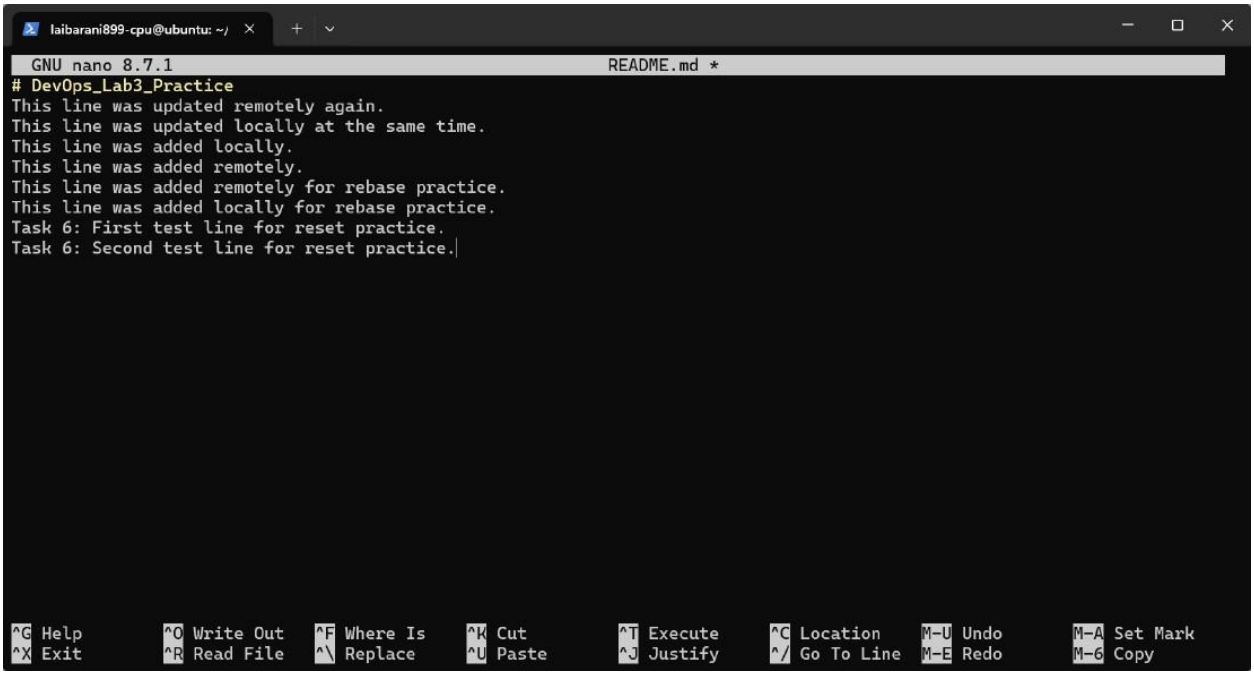
Task 6: Resetting Commits (Soft vs Hard Reset)



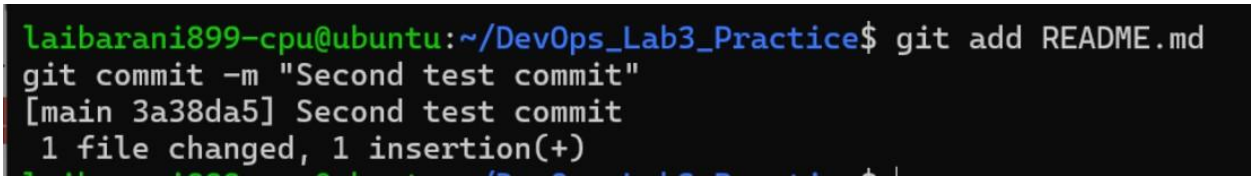
First test line added to README.md.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added test line"
[main fdfd06d] Added test line
 1 file changed, 1 insertion(+)
```

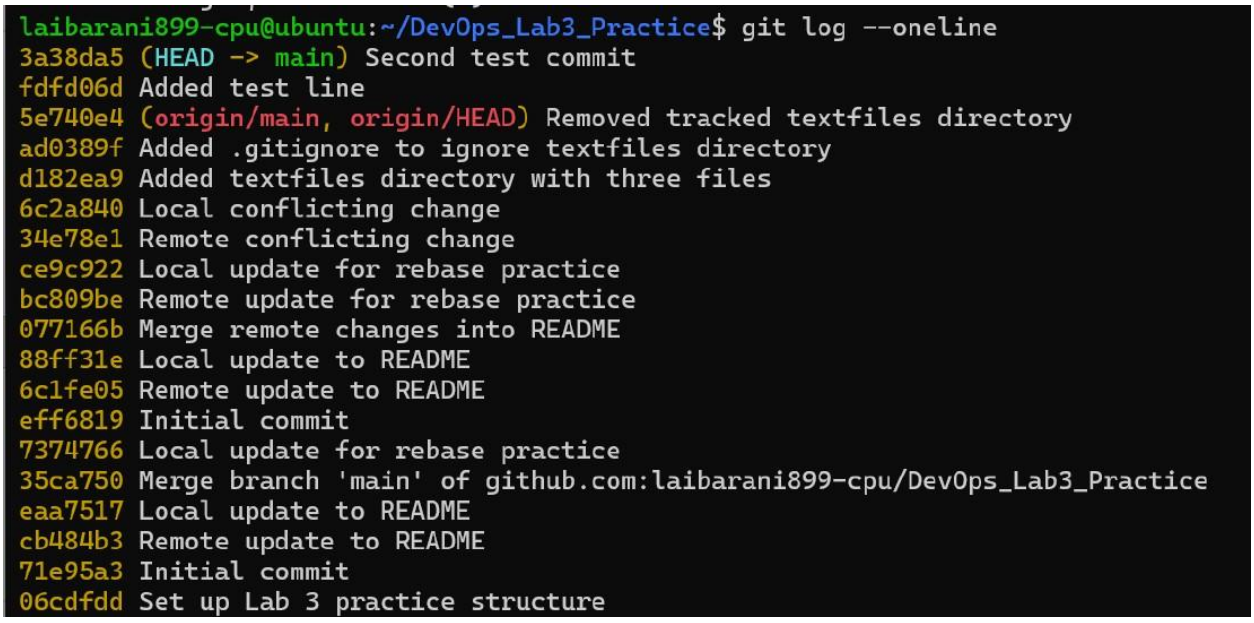
First commit created (“Added test line”).



Second test line added to README.md.



Second commit created (“Second test commit”).



Commit history before the reset.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice
This line was updated remotely again.
This line was updated locally at the same time.
This line was added locally.
This line was added remotely.
This line was added remotely for rebase practice.
This line was added locally for rebase practice.
Task 6: First test line for reset practice.
Task 6: Second test line for reset practice.
```

README.md before the reset; both test lines are present.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$
```

Soft reset performed with git reset—soft HEAD~1.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
fdfd06d (HEAD -> main) Added test line
5e740e4 (origin/main, origin/HEAD) Removed tracked textfiles directory
ad0389f Added .gitignore to ignore textfiles directory
d182ea9 Added textfiles directory with three files
6c2a840 Local conflicting change
34e78e1 Remote conflicting change
ce9c922 Local update for rebase practice
bc809be Remote update for rebase practice
077166b Merge remote changes into README
88ff31e Local update to README
6c1fe05 Remote update to README
eff6819 Initial commit
7374766 Local update for rebase practice
35ca750 Merge branch 'main' of github.com:laibarani899-cpu/DevOps_Lab3_Practice
eaa7517 Local update to README
cb484b3 Remote update to README
71e95a3 Initial commit
06cdfdd Set up Lab 3 practice structure
```

Commit history after the soft reset; the second commit is gone.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

git status after the soft reset; the changes are still staged.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ tail -n 5 README.md
This line was added remotely.
This line was added remotely for rebase practice.
This line was added locally for rebase practice.
Task 6: First test line for reset practice.
Task 6: Second test line for reset practice.
```

README.md after the soft reset; both lines are still present.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second Test commit"
[main 473a80e] Second Test commit
 1 file changed, 1 insertion(+)
```

Changes committed again after the soft reset ("Second Test commit").

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at fdfd06d Added test line
```

Hard reset performed with git reset—hard HEAD~1.

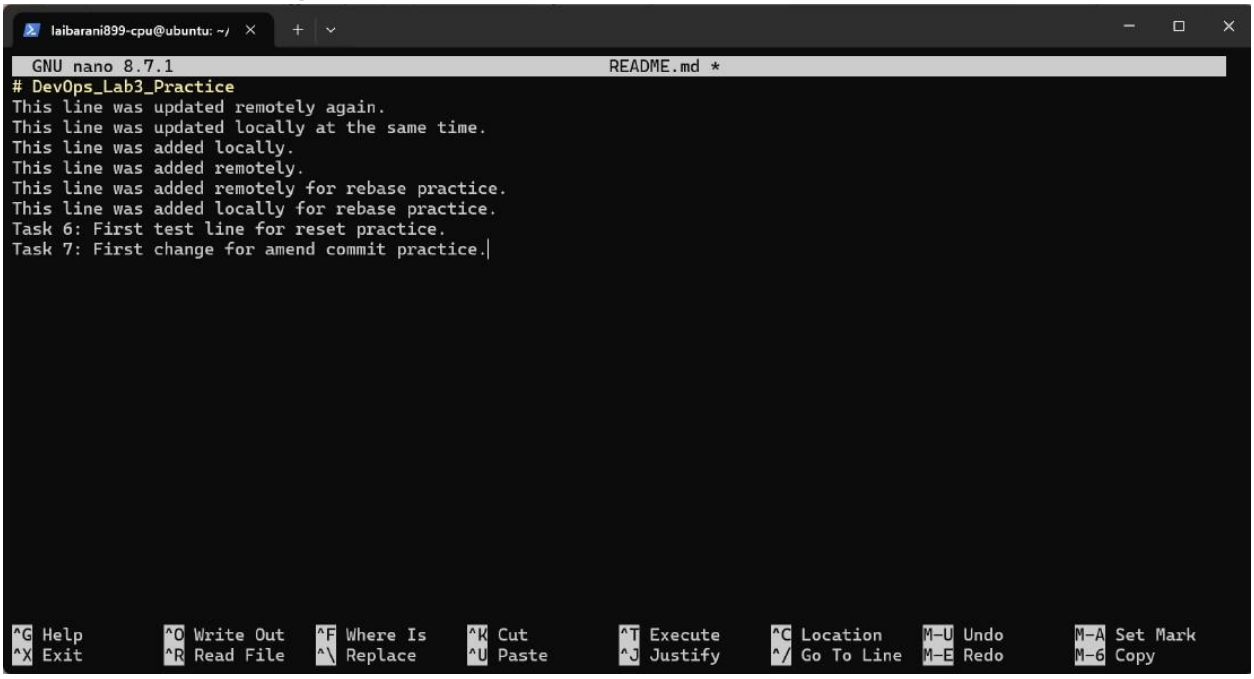
```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
fdfd06d (HEAD -> main) Added test line
5e740e4 (origin/main, origin/HEAD) Removed tracked textfiles directory
ad0389f Added .gitignore to ignore textfiles directory
d182ea9 Added textfiles directory with three files
6c2a840 Local conflicting change
34e78e1 Remote conflicting change
ce9c922 Local update for rebase practice
bc809be Remote update for rebase practice
077166b Merge remote changes into README
88ff31e Local update to README
6c1fe05 Remote update to README
eff6819 Initial commit
7374766 Local update for rebase practice
35ca750 Merge branch 'main' of github.com:laibarani899-cpu/DevOps_Lab3_Practice
eaa7517 Local update to README
cb484b3 Remote update to README
71e95a3 Initial commit
06cdfdd Set up Lab 3 practice structure
```

Commit history after the hard reset.

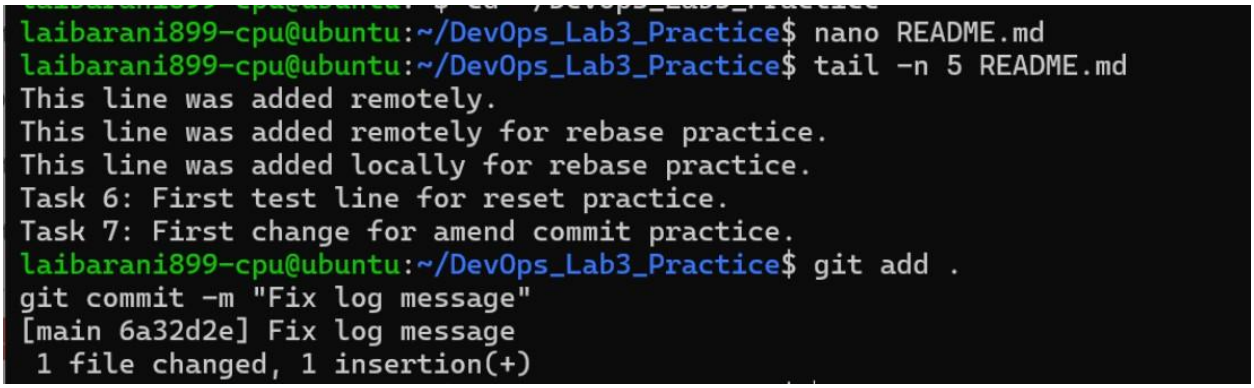
```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ tail -n 5 README.md
This line was added locally.
This line was added remotely.
This line was added remotely for rebase practice.
This line was added locally for rebase practice.
Task 6: First test line for reset practice.
```

README.md after the hard reset; the second test line is removed.

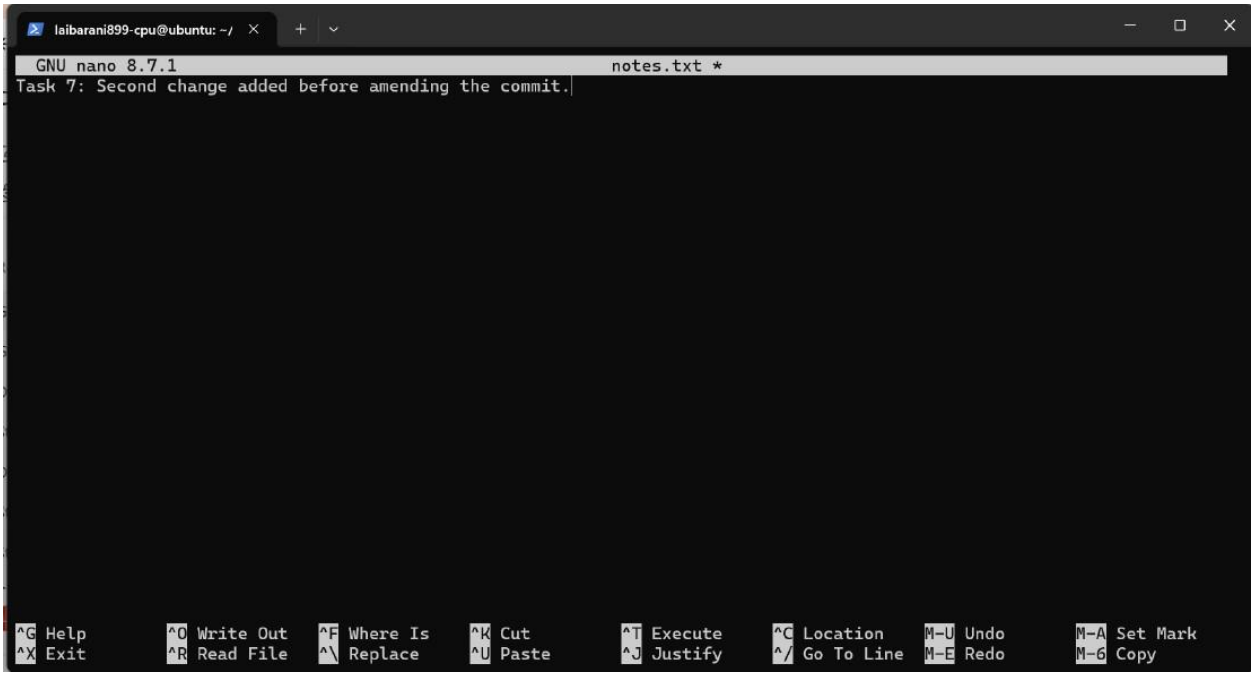
Task 7: Amending the Last Commit



README.md edited for the amend task.



Change committed with the message “Fix log message”.



Second file (notes.txt) created, which was forgotten in the first commit.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ nano notes.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add .
git commit --amend
[main 59a5403] Fix log message
 Date: Sat Oct 17 16:05:05 2026 +0000
 2 files changed, 2 insertions(+)
 create mode 100644 notes.txt
```

git commit—amend adds notes.txt to the previous commit (2 files changed).

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ echo "temporary text" >> temp.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add .
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added temporary text file"
[main 8eff45e] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -5
8eff45e (HEAD -> main) Added temporary text file
59a5403 Fix log message
fdfd06d Added test line
5e740e4 (origin/main, origin/HEAD) Removed tracked textfiles directory
ad0389f Added .gitignore to ignore textfiles directory
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git switch -c task8-safe-revert origin/main
branch 'task8-safe-revert' set up to track 'origin/main'.
Switched to a new branch 'task8-safe-revert'
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git cherry-pick 8eff45e
[task8-safe-revert 5ddc493] Added temporary text file
 Date: Sat Oct 17 16:12:46 2026 +0000
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git cherry-pick 8eff45e
On branch task8-safe-revert
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

You are currently cherry-picking commit 8eff45e.
  (all conflicts fixed: run "git cherry-pick --continue")
  (use "git cherry-pick --skip" to skip this patch)
  (use "git cherry-pick --abort" to cancel the cherry-pick operation)

nothing to commit, working tree clean
The previous cherry-pick is now empty, possibly due to conflict resolution.
If you wish to commit it anyway, use:

    git commit --allow-empty

Otherwise, please use 'git cherry-pick --skip'
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git cherry-pick --skip
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch task8-safe-revert
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin task8-safe-revert:main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 75.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   5e740e4..5ddc493  task8-safe-revert -> main
```

temp.txt created and committed; the commit was applied on the task8-safe-revert branch with cherry-pick and pushed to the remote main.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -5
5ddc493 (HEAD -> task8-safe-revert, origin/main, origin/HEAD) Added temporary text file
5e740e4 Removed tracked textfiles directory
ad0389f Added .gitignore to ignore textfiles directory
d182ea9 Added textfiles directory with three files
6c2a840 Local conflicting change
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ ^C
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git revert 5ddc493
[task8-safe-revert 151aaf9] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt
```

git revert creates a new commit that undoes “Added temporary text file” (temp.txt deleted).

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin task8-safe-revert:main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 259 bytes | 86.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
   5ddc493..151aaf9  task8-safe-revert -> main
```

Revert commit pushed to the remote main branch.

Task 9: Force Push (With Caution)

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b test-force
Switched to a new branch 'test-force'
```

New branch test-force created.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ echo "Force push practice" > force.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git add force.txt
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Add force push practice file"
[test-force cc75981] Add force push practice file
 1 file changed, 1 insertion(+)
 create mode 100644 force.txt
```

force.txt created and committed on test-force.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin test-force
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 50.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/laibarani899-cpu/DevOps_Lab3_Practice/pull/new/test-force
remote:
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
 * [new branch]      test-force -> test-force
branch 'test-force' set up to track 'origin/test-force'.
```

test-force branch pushed to the remote.

```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 151aaf9 Revert "Added temporary text file"
```

git reset—hard HEAD~1 removes the latest local commit.

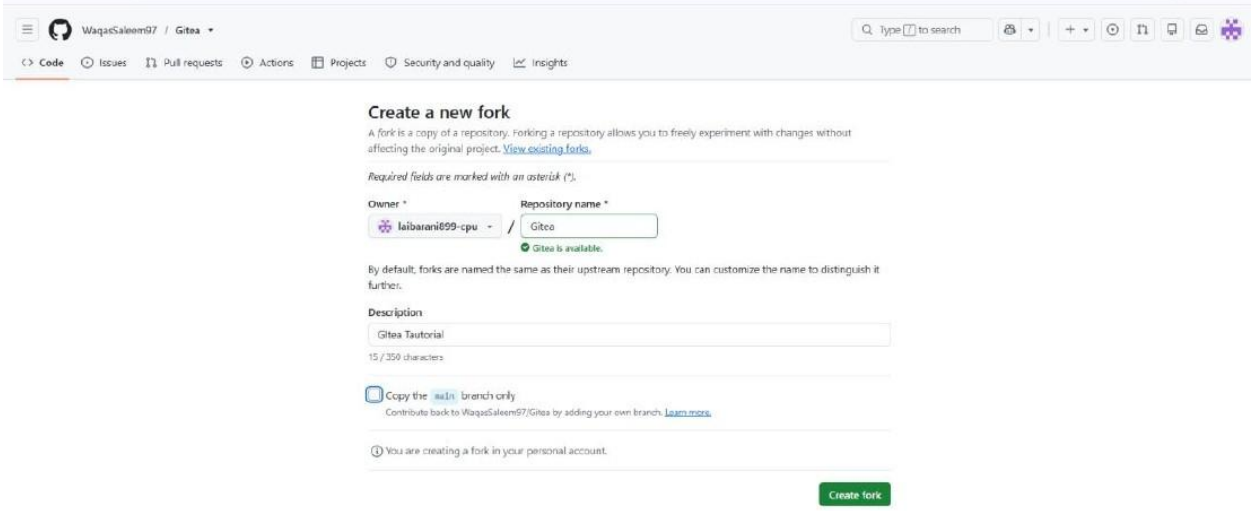
```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:laibarani899-cpu/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

Normal push rejected (non-fast-forward) because the histories differ.

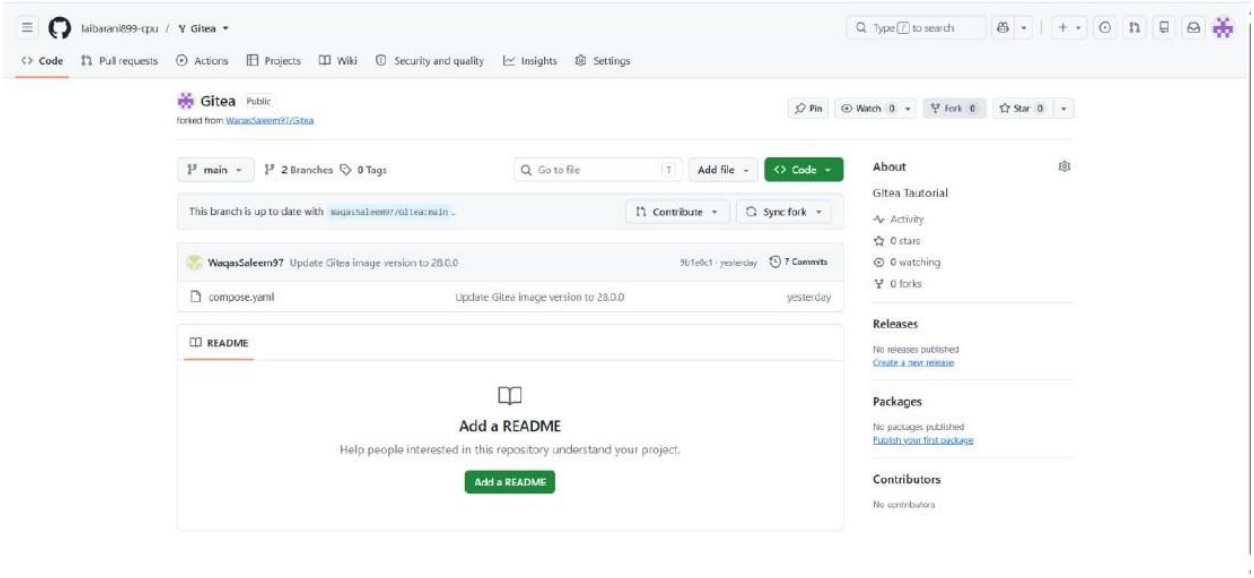
```
laibarani899-cpu@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:laibarani899-cpu/DevOps_Lab3_Practice.git
 + cc75981...151aaf9 test-force -> test-force (forced update)
```

git push—force overwrites the remote branch history (forced update).

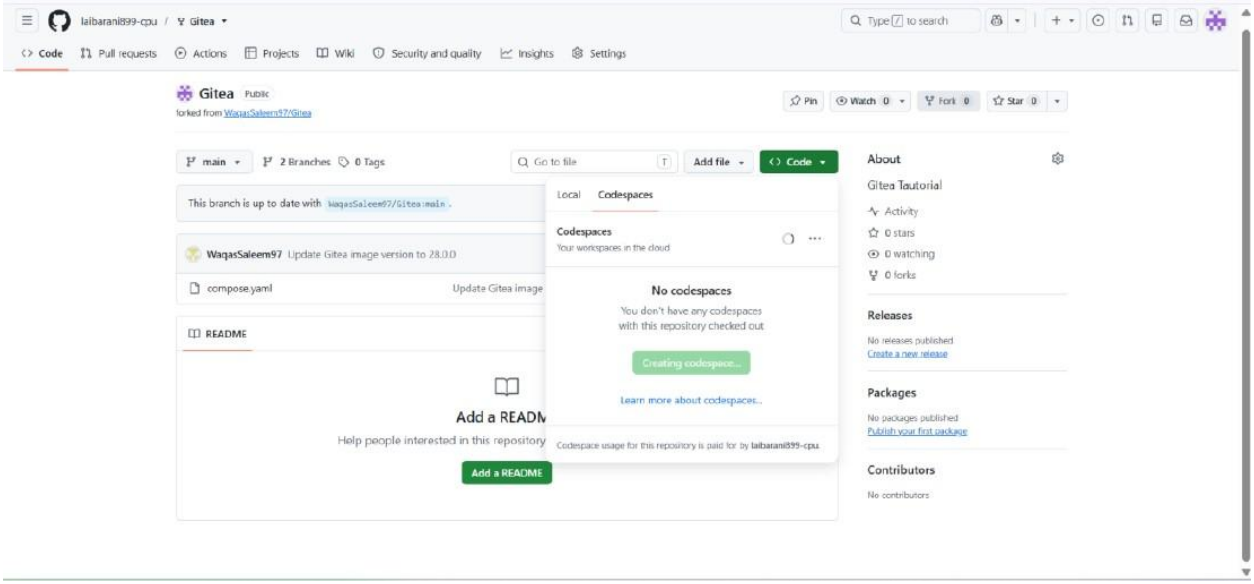
Task 10: Running Gitea in GitHub Codespaces via Docker Compose



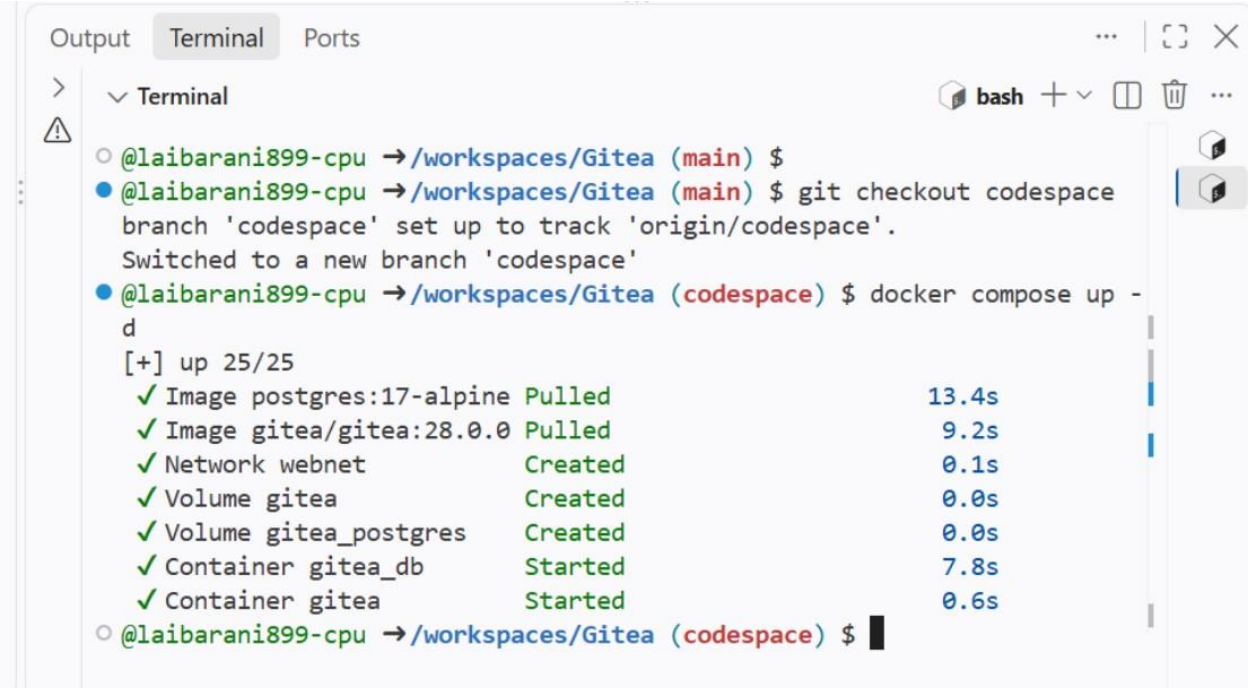
Creating the fork of the Gitea repository (“Copy the main branch only” unchecked).



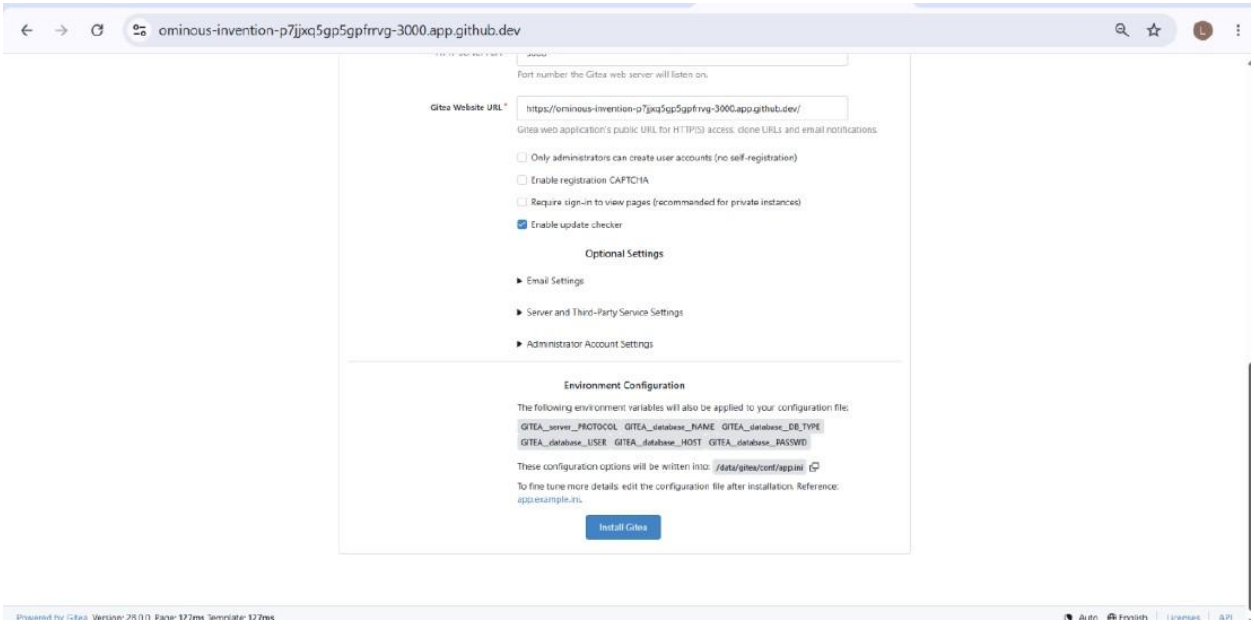
Forked Gitea repository in my GitHub account.



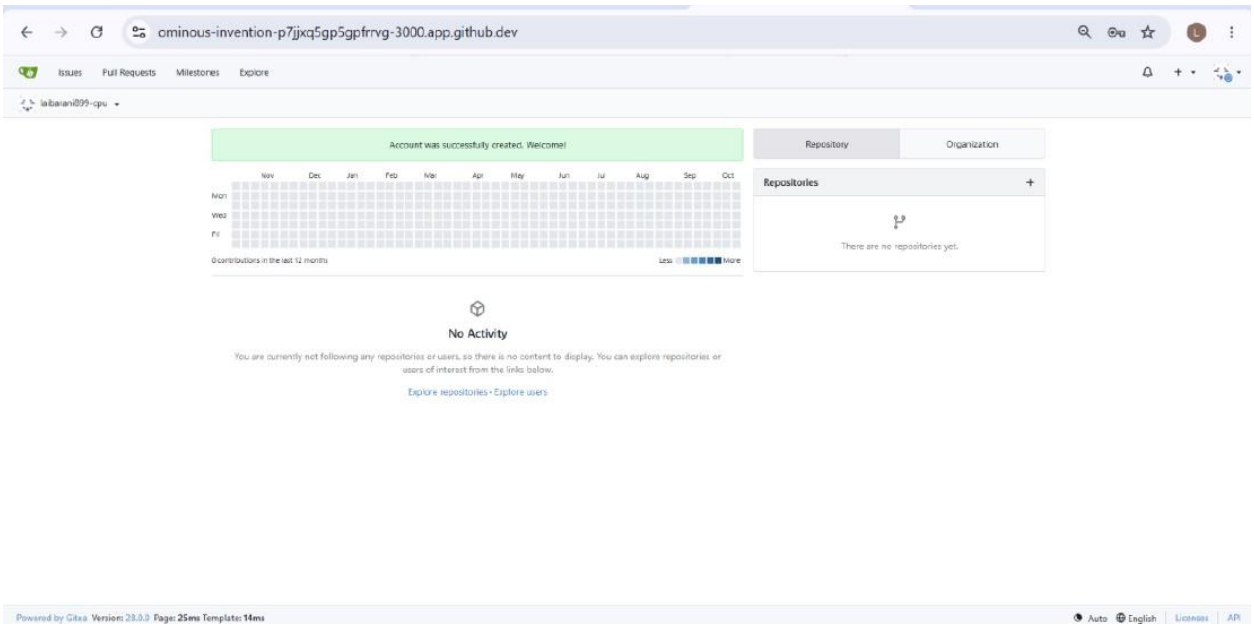
Codespace being created from the forked repository.



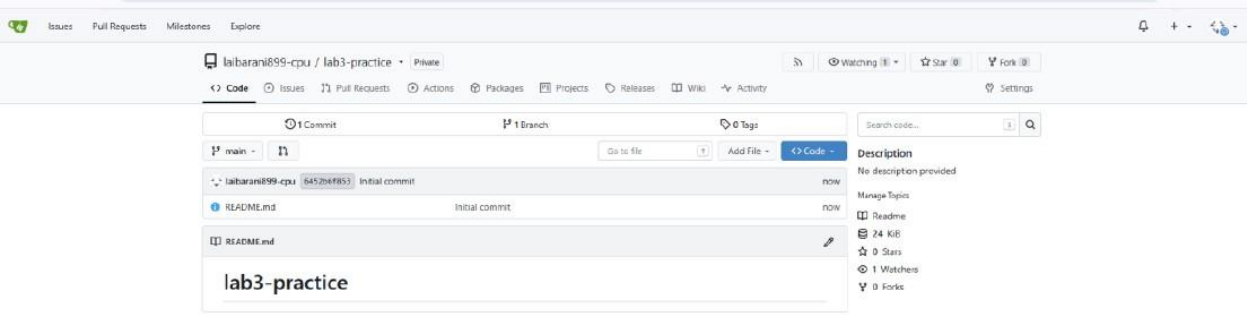
Switched to the codespace branch and started Gitea with docker compose up -d; containers running.



Gitea installation page opened on forwarded port 3000.

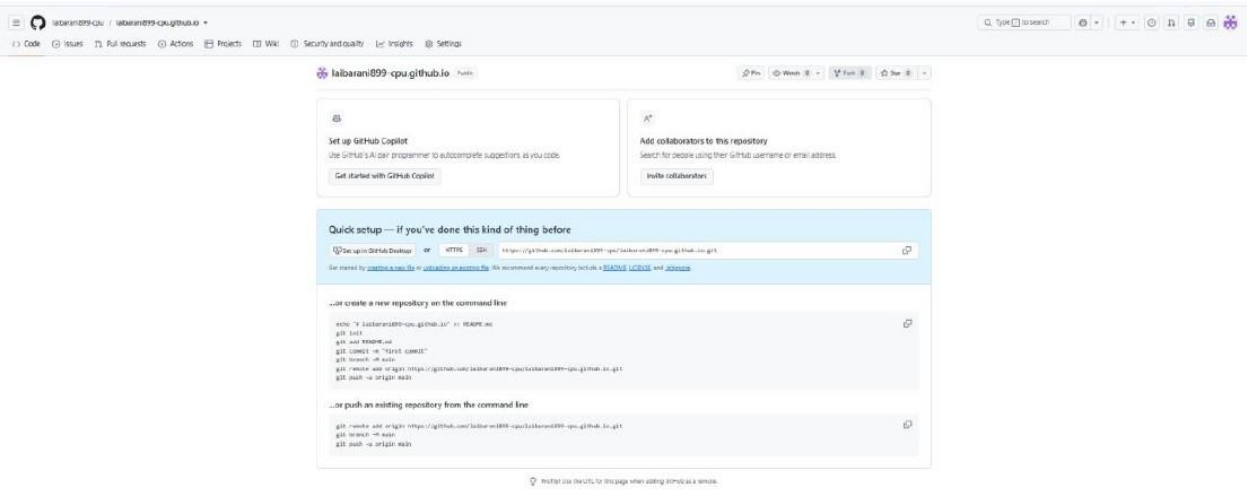


Gitea dashboard after the admin account was created and logged in.

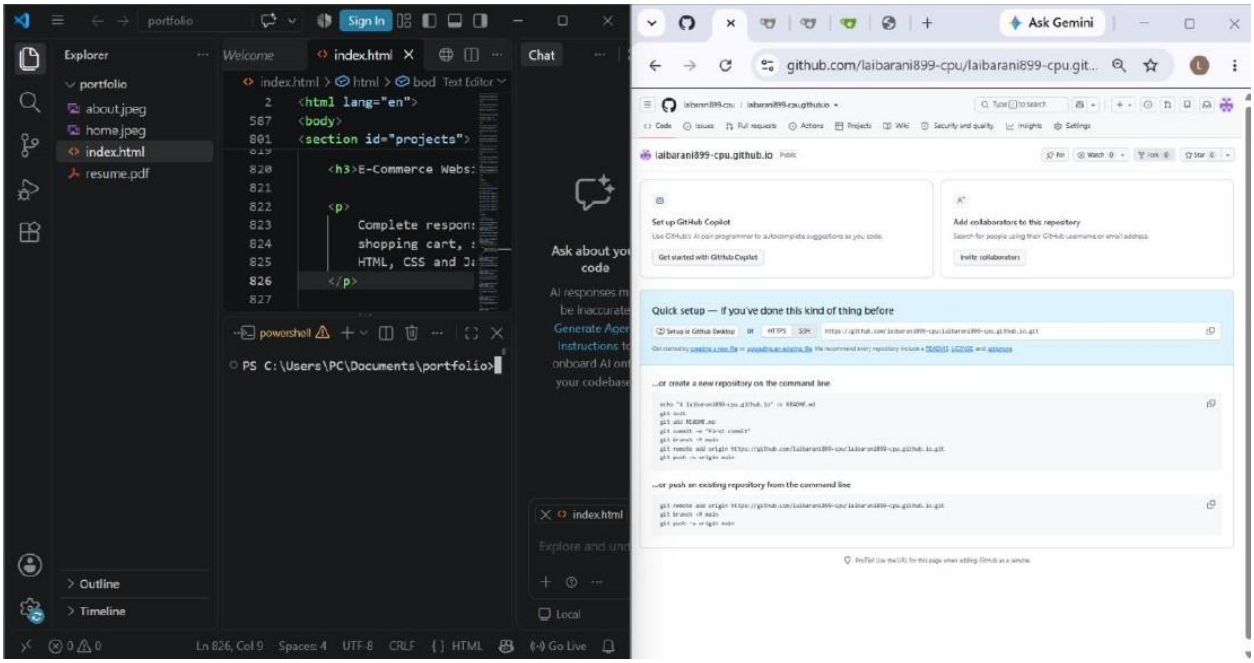


New repository (lab3-practice) created in Gitea.

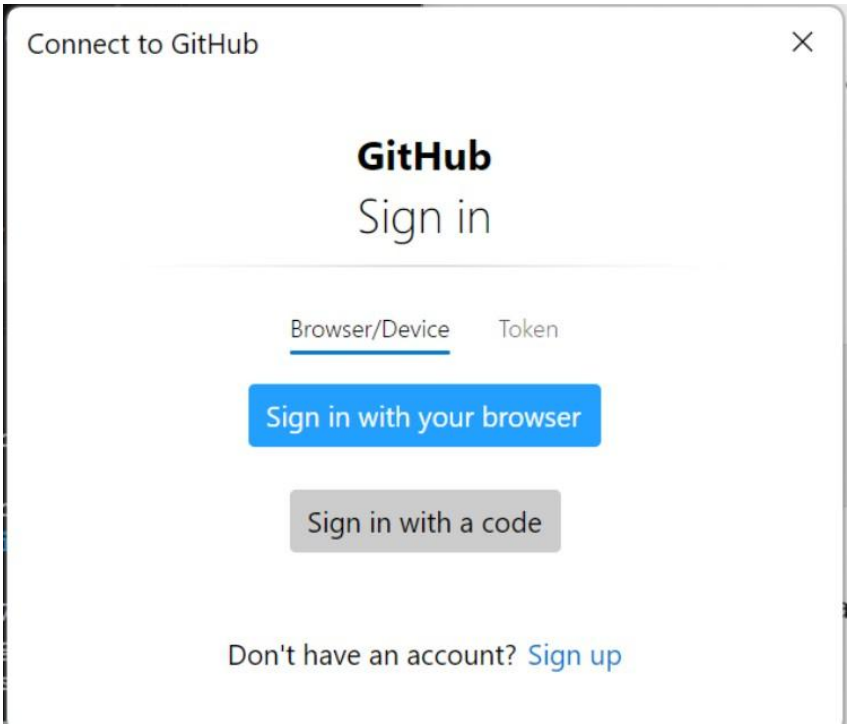
Task 11: Creating a GitHub Pages Portfolio Site



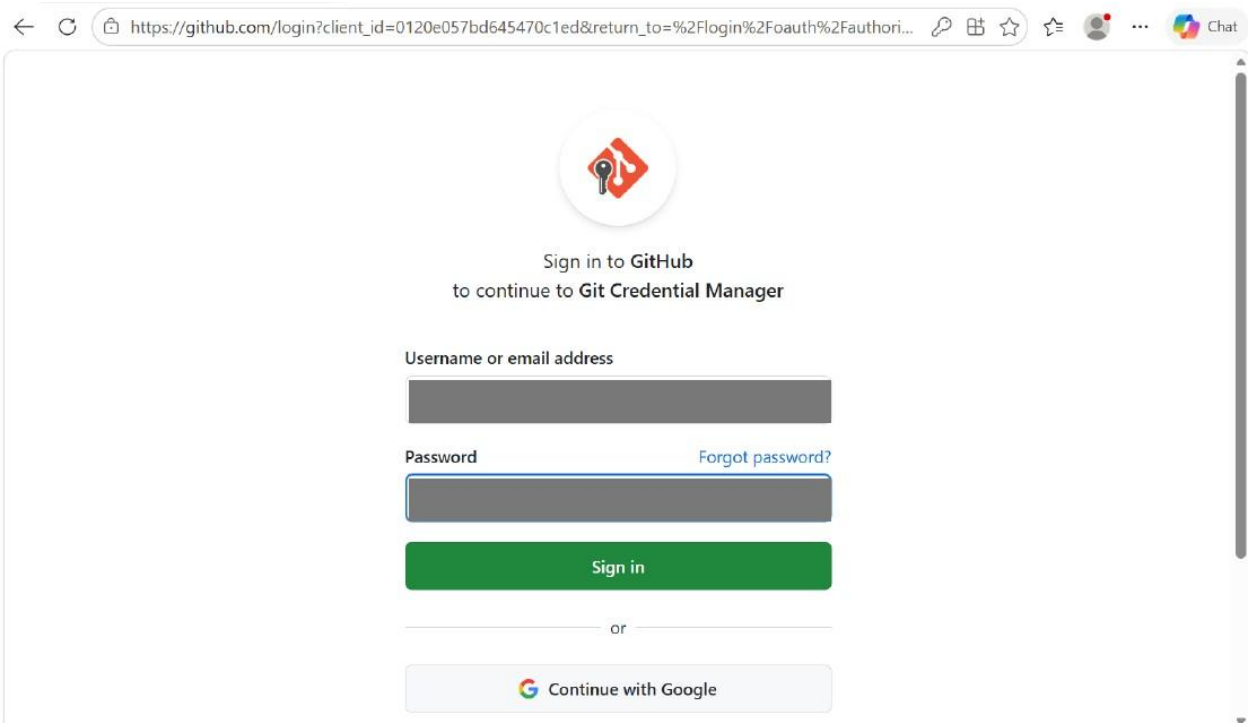
New public repository laibarani899-cpu.github.io created on GitHub.



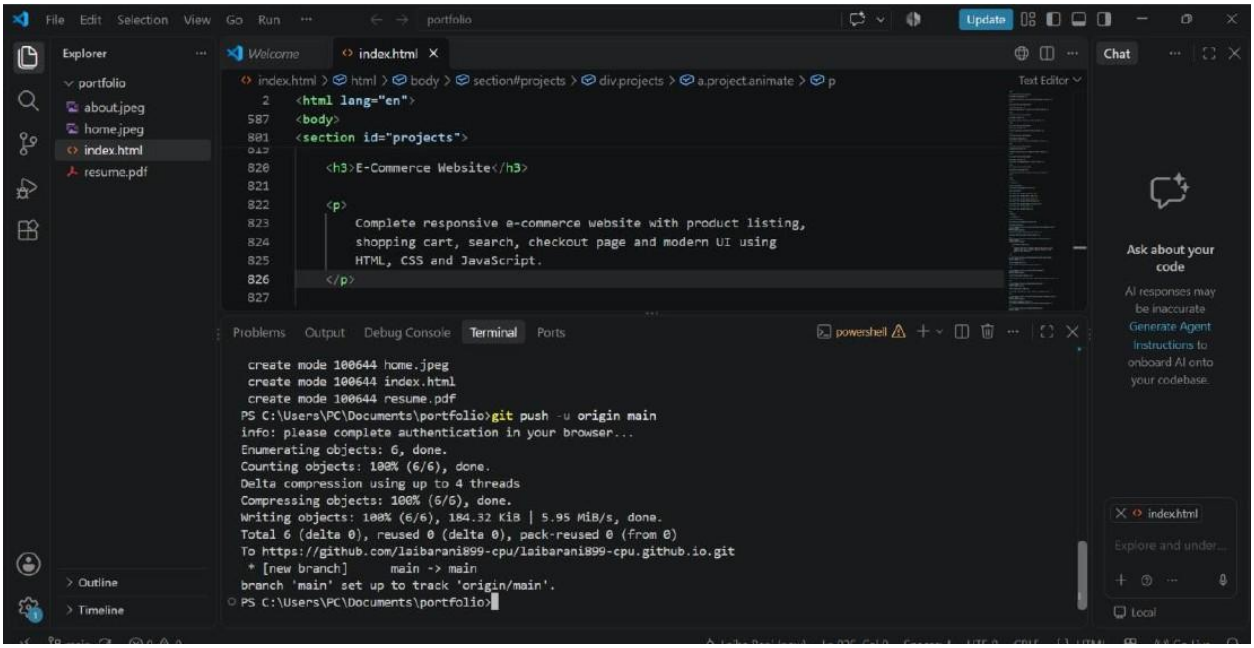
Portfolio files (index.html, images, resume.pdf) in the local folder.



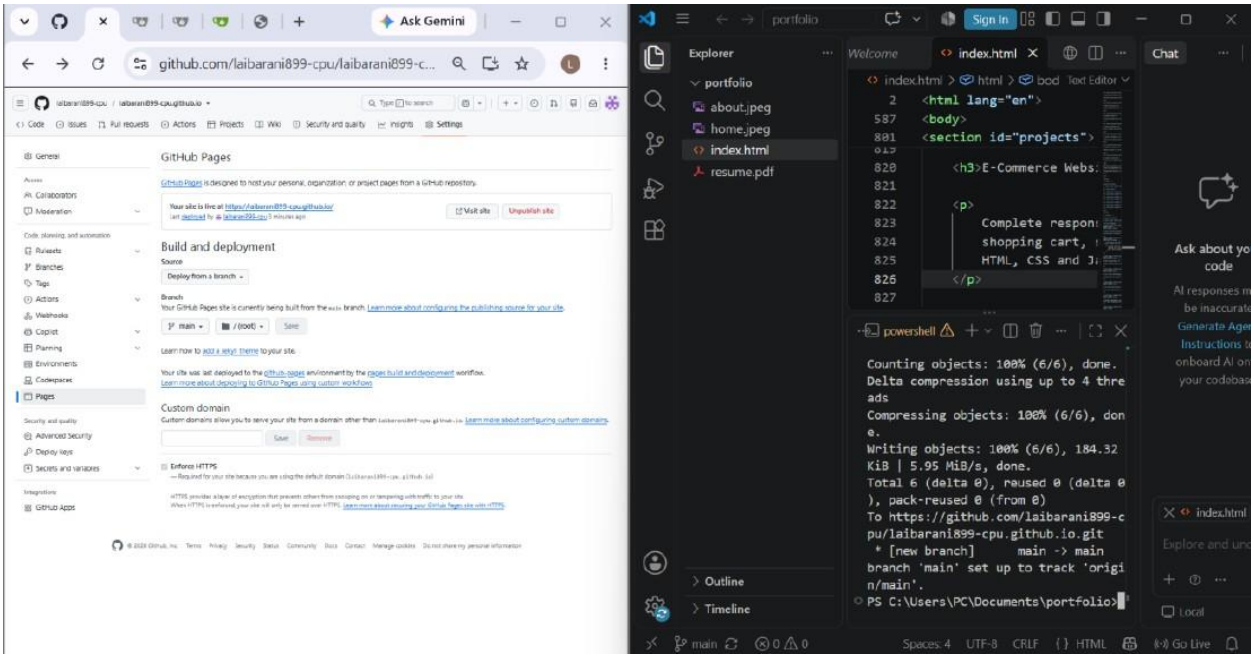
GitHub sign-in prompt shown while pushing the files.



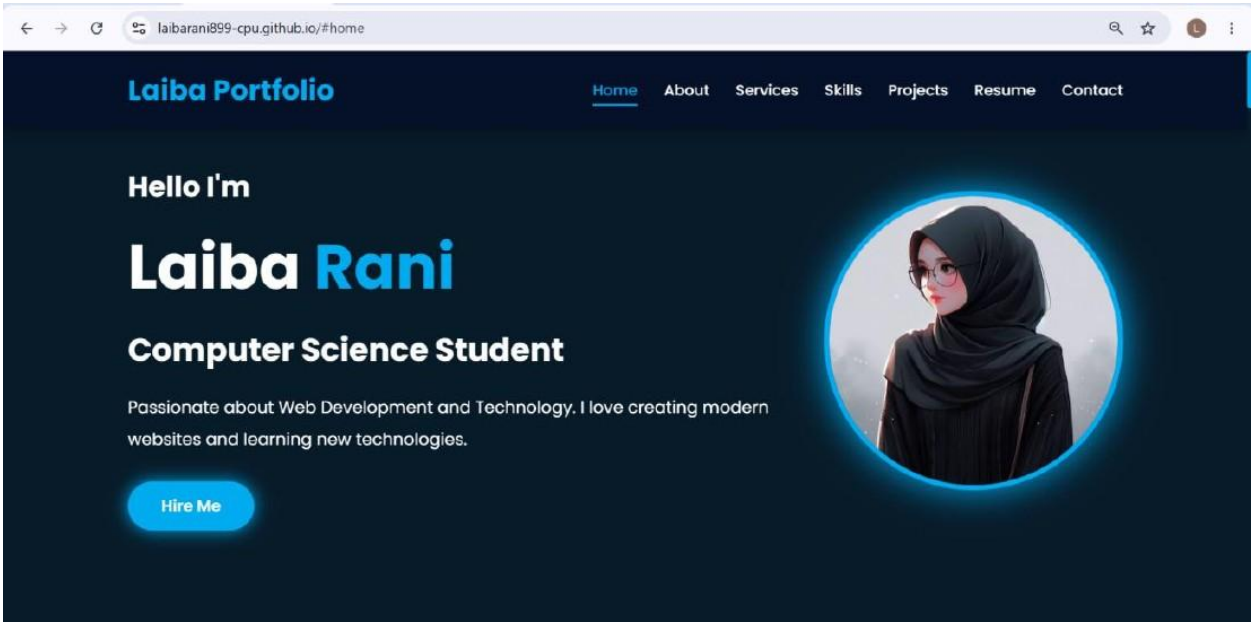
Browser authentication through Git Credential Manager (credentials hidden).



Portfolio files pushed successfully to the GitHub repository.



GitHub Pages settings showing the site is published.



Live portfolio site opened in the browser.